Lesson 4.1 - Quartiles, Deciles, and Percentiles

> **Video:** Quartiles, Deciles, & Percentiles With Cumulative Relative Frequency by The Organic Chemistry Tutor **Chapter:** 4 - Measures of Position

What they are

Measure	Splits the sorted data into	Memory aid
Quartiles Q_1, Q_2, Q_3	**4** equal parts	4 quarters = 1 dollar
Deciles $D_1, \ldots, D_9$	**10** equal parts	"deci" = one tenth (10 decimeters = 1 meter)
Percentiles $P_1, \ldots, P_{99}$	**100** equal parts	per-**cent**

They line up with each other:

```
Percentile:  0   10   20   25   30   40   50   60   70   75   80   90  100
Decile:      D0  D1   D2        D3   D4   D5   D6   D7        D8   D9  D10
Quartile:    min           Q1             Q2                  Q3         max
```

- $Q_1 = P_{25}, \quad Q_2 = D_5 = P_{50} = \text{median}, \quad Q_3 = P_{75}$
- $D_k = P_{10k}$, so $D_4 = P_{40}$ and $D_9 = P_{90}$.

What does a percentile mean?

The **70th percentile** is a value such that **70%** of the data is **at or below** it and **30%** is **at or above** it.

Example: if your SAT score is at the 70th percentile, you scored as well as or better than 70% of test-takers, and 30% scored as well as or better than you.

Method 1 - quartiles as medians of halves

- Q_2 = median of the whole data set.
- Q_1 = median of the **lower half**; Q_3 = median of the **upper half**. When n is odd, the median is left out of both halves.

Example (odd $n = 11$): 2, 3, 5, 7, 8, **10**, 11, 13, 15, 16, 19

- $Q_2 = 10$
- Lower half 2, 3, **5**, 7, 8 → $Q_1 = 5$
- Upper half 11, 13, **15**, 16, 19 → $Q_3 = 15$

Method 2 - the percentile location formula

The **position** of the k-th percentile in the sorted list is

$$L_k = \frac{k}{100}(n+1)$$

- If L_k is a whole number, the percentile is the value at that position.
- If L_k falls between two positions, the video **averages the two neighboring values**.

Same data ($n = 11$):

Percentile	Location	Value
$P_{25} = Q_1$	$\frac{25}{100}(12) = 3$ → 3rd value	5
$P_{50} = Q_2$	$\frac{50}{100}(12) = 6$ → 6th value	10
$P_{75} = Q_3$	$\frac{75}{100}(12) = 9$ → 9th value	15

Both methods agree.

Example with even $n = 16$

The video uses a list of 16 numbers shown on screen. The positions that matter are:

Position	4	5	6	8	9	10	11	12	13
Value	9	11	12	15	16	16	17	19	20

- **Medians of halves:** $Q_2 = \dfrac{15+16}{2} = 15.5,\ Q_1 = \dfrac{9+11}{2} = 10,\ Q_3 = \dfrac{19+20}{2} = 19.5$
- **Formula for Q_1:** $L_{25} = \frac{k}{100}(n+1) = \frac{25}{100}(17) = 4.25$, between the 4th and 5th values → $\dfrac{9+11}{2} = 10$ ✓
- **Formula for Q_3:** $L_{75} = \frac{k}{100}(n+1) = \frac{75}{100}(17) = 12.75$, between the 12th and 13th values → $\dfrac{19+20}{2} = 19.5$ ✓
- **6th decile (P_{60}):** $L_{60} = \frac{k}{100}(n+1) = \frac{60}{100}(17) = 10.2$, between the 10th (16) and 11th (17) values → $\dfrac{16+17}{2} = 16.5$

> **About D_6:** the location 10.2 is much closer to position 10 than to 11, so the video suggests the true answer should be closer to 16. **Linear interpolation** makes this precise: $16 + 0.2(17-16) = 16.2$. With small n these estimates are rough, and they become more accurate as n grows.

Going backwards - which percentile is a given value?

$$\text{percentile of value } v = \frac{x + 0.5\,y}{n} \times 100$$

- x = number of values **less than** v
- y = number of times v appears
- n = total number of values

Using the same 16-number list:

- **Value 12:** 5 values are below it and 12 appears once.

  $$\frac{5 + 0.5(1)}{16} \times 100 = 34.375 \approx 34\text{th percentile}$$

  This makes sense: 12 lies between Q_1 (25th) and Q_2 (50th).

- **Value 16:** 8 values are below it and 16 appears twice.

$$\frac{8 + 0.5(2)}{16} \times 100 = 56.25 \approx 56\text{th percentile}$$

Percentiles from a cumulative relative frequency table

Data (20 values):

Value	f	RF	CRF
2	2	0.10	0.10
3	4	0.20	0.30
4	3	0.15	0.45
6	2	0.10	0.55
7	1	0.05	0.60
9	5	0.25	0.85
10	2	0.10	0.95
12	1	0.05	1.00
Total	**20**	**1.00**	

Rules:

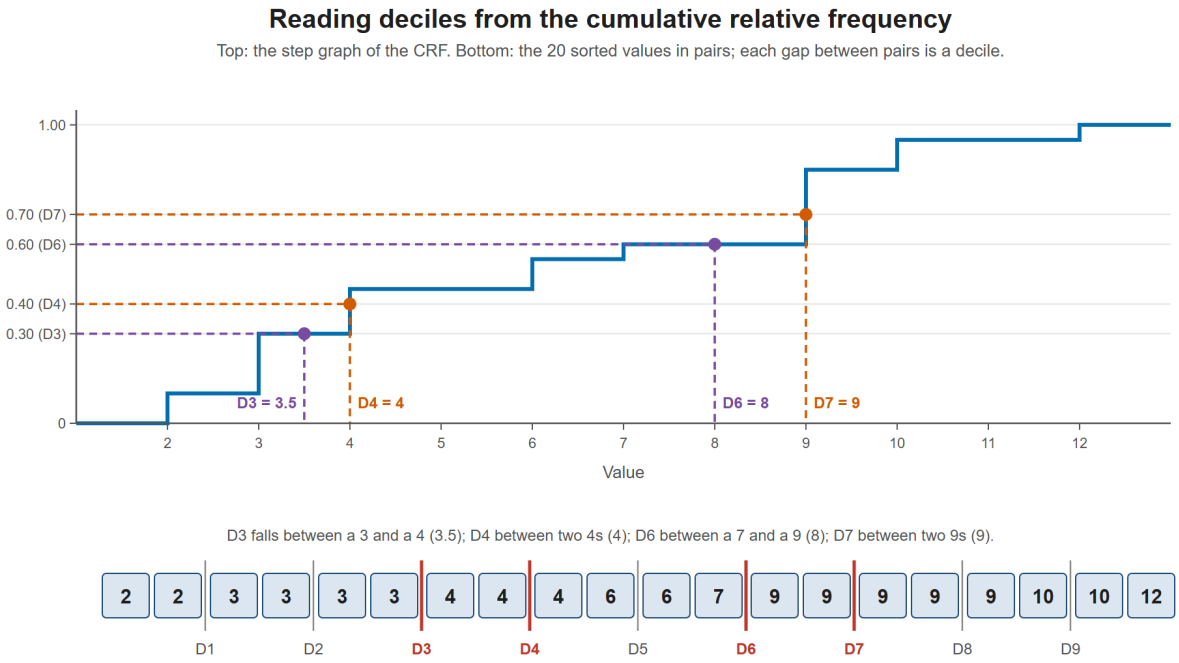
- If the percentile falls **strictly between** two CRF values, take the value of the **higher** row.
- If it equals a CRF value **exactly**, average that row's value with the next row's value.

Decile	Percentile	Where it falls	Value
D_4	40th	between 0.30 and 0.45	**4**
D_7	70th	between 0.60 and 0.85	**9**
D_3	30th	exactly 0.30 → average 3 and 4	**3.5**
D_6	60th	exactly 0.60 → average 7 and 9	**8**

Check by listing the data in pairs (20 values ÷ 10 deciles = 2 values per decile):

```
2 2 | 3 3 | 3 3 | 4 4 | 4 6 | 6 7 | 9 9 | 9 9 | 9 10 | 10 12
   D1    D2    D3    D4    D5    D6    D7    D8     D9
```

- D_3 sits between a 3 and a 4 → 3.5
- D_4 sits between two 4s → 4
- D_6 sits between a 7 and a 9 → 8
- D_7 sits between two 9s → 9



Reading deciles from the cumulative relative frequency. Where a decile level crosses a flat step, it lands strictly between two values, so take the higher one (D4 = 4, D7 = 9). Where it lands exactly on a step, average the two values on either side (D3 = 3.5, D6 = 8). The row of boxes below shows the same thing with the 20 sorted values.

Key takeaways

- Quartiles split the data into 4 parts, deciles into 10, percentiles into 100. $Q_2 = D_5 = P_{50}$ = median.
- Location of P_k: $\frac{k}{100}(n+1)$. If it is not a whole number, use the neighboring values (average them, or interpolate).
- Percentile of a value: $\frac{x + 0.5y}{n} \times 100$.
- From a CRF table: strictly between two rows → higher value; exactly on a row → average with the next value.
- Different textbooks and software use slightly different percentile rules. Small differences are normal.